# Differential Equations

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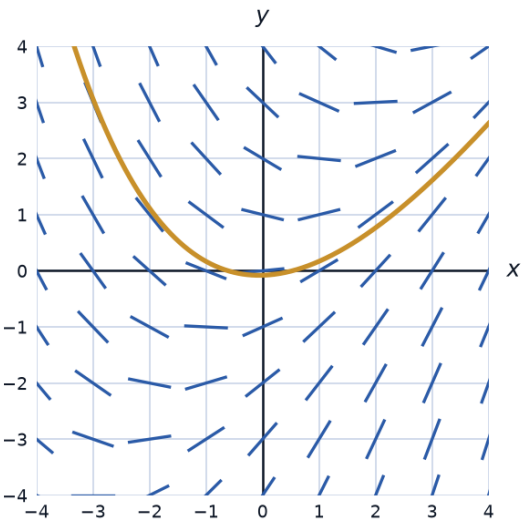
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$.

**2** [Foundational] • Target: Forming differential equations from a description

The rate of decrease of a quantity $Q$ is proportional to $Q$ itself. Write a differential equation for $\frac{dQ}{dt}$, using $k > 0$ as the constant of proportionality.




**3** [Foundational] • Target: Forming differential equations from a description

A tank drains so that the rate of decrease of its volume $V$ is proportional to $\sqrt{V}$. Write a differential equation for $\frac{dV}{dt}$, using $k > 0$.

**4** [Foundational] • Target: Forming differential equations from a description

A chemical reaction's concentration $C$ decreases at a rate proportional to the square of the concentration. Write a differential equation for $\frac{dC}{dt}$, using $k > 0$.

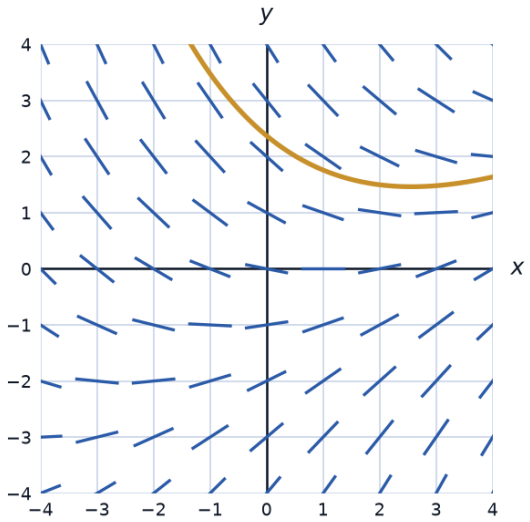
**5** [Foundational] • Target: Solving modeling equations with initial conditions

A population $P$ satisfies $\frac{dP}{dt} = 0.08P$. If $P = 500$ when $t = 0$, find $P$ when $t = 10$, to the nearest whole number.

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**6** [Foundational] • Target: Solving modeling equations with initial conditions

Solve $\frac{dV}{dt} = -k\sqrt{V}$ (from the tank problem above), given $V(0) = V_0$.




**7** [Foundational] • Target: First-order linear differential equations using an integrating factor

Solve $\frac{dy}{dx} + 3y = 12$ using an integrating factor, given $y = 2$ when $x = 0$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = 3e^{2x}$ satisfies $\frac{dy}{dx} - 2y = 0$.

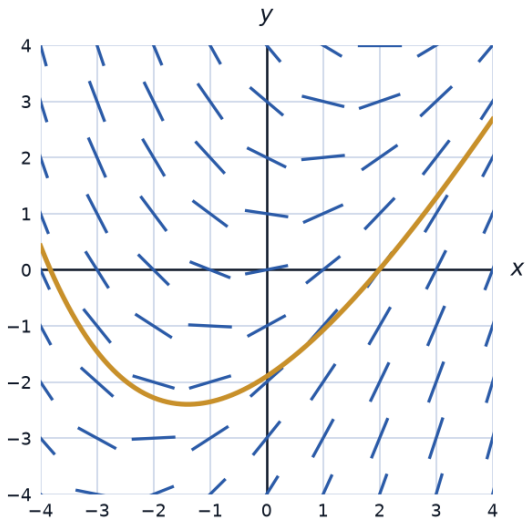
**9** [Intermediate] • Target: Verifying solutions to differential equations

Verify that $y = x^2 + Ce^{-x}$ is a general solution of $\frac{dy}{dx} + y = x^2 + 2x$.

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**10** [Intermediate] • Target: General and particular solutions

Find the general solution of $\frac{dy}{dx} = 4x^3 - 6x$, and explain what determines a particular solution from this family.




**11** [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = 6x^2 - 4$ and passes through $(1, -1)$. Find the equation of this particular solution.

**12** [Intermediate] • Target: General and particular solutions

A curve satisfies $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ and passes through $(1, 4)$. Find the particular solution.

**13** [Intermediate] • Target: Applications: Newton's law of cooling

A cup of tea at $85\,^\circ$C is placed in a room at $22\,^\circ$C. Using $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 8 minutes the tea has cooled to $65\,^\circ$C, find $k$, to three decimal places.

**14** [Intermediate] • Target: Applications: Newton's law of cooling

Using the value of $k$ found above, find the temperature of the tea after 20 minutes, to the nearest degree.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

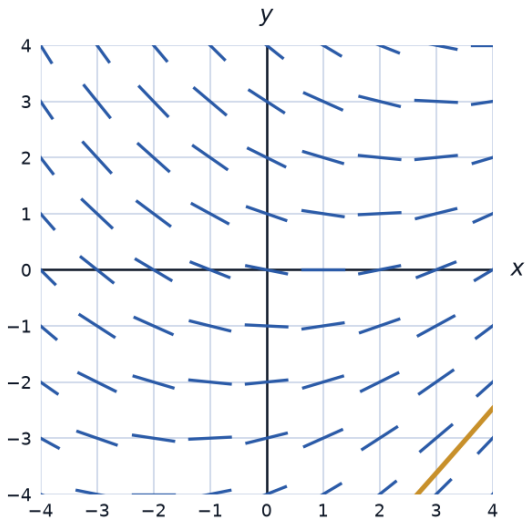
**15** [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 12.01 Poisson Distribution

A metal rod at $200\,^\circ$C cools in air at $25\,^\circ$C, with cooling constant $k = 0.03$ per minute. Find its temperature after 15 minutes, to the nearest degree.

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**16** [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 13.01 Continuous Random Variables

This question has three parts. A tank contains $300$ litres of brine with $10$ kg of dissolved salt. Fresh water flows in at $6$ litres/minute, and the well-mixed solution flows out at the same rate, keeping the volume constant. Let $S(t)$ be the mass of salt (kg) at time $t$ minutes. (a) Explain why $\frac{dS}{dt} = -\frac{S}{50}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 10$. (c) Find how long it takes for the salt content to drop to $2$ kg, to the nearest minute.




**17** [Challenging] • Target: Applications: Newton's law of cooling • Cross-Lesson: 14.01 Sampling & Estimation

The temperature $\theta$ of a cooling metal rod satisfies $\frac{d\theta}{dt} = -k(\theta - 20)$, where $t$ is time in minutes and $k$ is a constant. Given $\theta = 80$ when $t = 0$ and $\theta = 50$ when $t = 10$, find the value of $t$ when $\theta = 30$.

**18** [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 15.01 Hypothesis Testing

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

**19** [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 2.01 Exponentials & Logarithms

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

**20** [Challenging] • Target: Forming differential equations from a description • Cross-Lesson: 3.01 Advanced Trigonometry

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Solving by separation of variables

First-order linear ODE solution structure: general solution = homogeneous solution + ____

**a.** particular solution

**b.** another homogeneous solution

**c.** the initial condition

**d.** zero

**22** [Foundational] • Target: Forming differential equations from a description

A radioactive substance has half-life 2 years and initial mass 16g. Find the mass remaining after 4 years.

**a.** 4

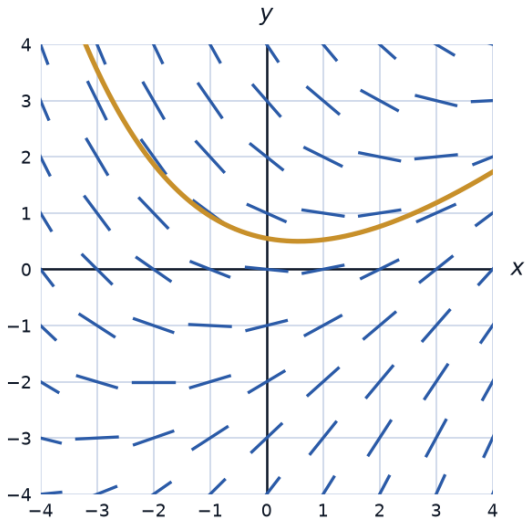
**b.** 16

**c.** 8

**d.** 5

**23** [Foundational] • Target: Solving modeling equations with initial conditions

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):




**a.** $y = \frac{1}{2}x^3 + Cx$

**b.** $y = \frac{1}{3}x^3 + C$

**c.** $y = x^3 + \frac{C}{x}$

**d.** $y = x^3 + Cx$

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**24** [Foundational] • Target: First-order linear differential equations using an integrating factor

Coffee at 70°C is placed in a 22°C room and cools to 45°C in 5 min (Newton's law of cooling). How long (min) until it reaches 35°C?

**a.** 7

**b.** $\frac{10\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

**c.** $\frac{5\ln(2)}{\ln\left(\frac{14}{9}\right)}$

**d.** $\frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

**25** [Foundational] • Target: Further integrating factor examples

Newton's law of cooling: $\frac{dT}{dt} = k(T - T_s)$, $k < 0$. Steady state:

**a.** $T = 0$

**b.** $T = kT_s$

**c.** $T = T_s$

**d.** $T = T_0$

**26** [Foundational] • Target: Verifying solutions to differential equations

A first-order linear ODE has the form:

**a.** $y' + P(x)y = Q(x)$

**b.** $y' = P(x)y^2$

**c.** $y'y = Q(x)$

**d.** $y'' = P(x)$

**27** [Foundational] • Target: General and particular solutions

For the linear ODE $y' + 2y = 4$: integrating factor $\mu =$

**a.** $e^2$

**b.** 2

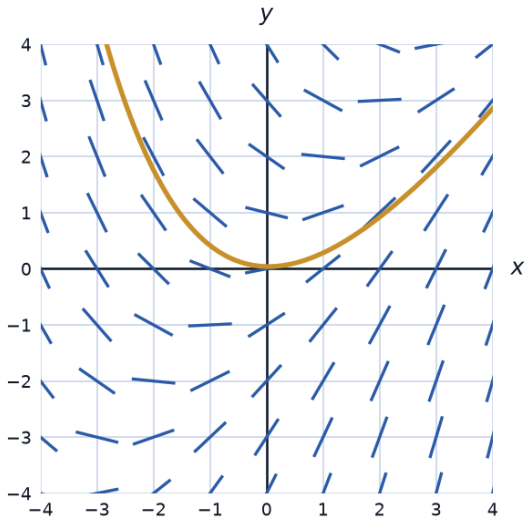
**c.** $e^{-2x}$

**d.** $e^{2x}$

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**28** [Foundational] • Target: Applications: Newton's law of cooling

Linear homogeneous 2nd-order ODE: if $y_1, y_2$ are solutions, so is:




**a.** $y_1^2$

**b.** $y_1 y_2$

**c.** $y_1'$

**d.** $y_1 + y_2$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: A well-designed word problem: a mixing tank

Which differential equation correctly models the statement: "The rate of increase of a population P is proportional to P itself."?

**a.** $\frac{dP}{dt} = kP$

**b.** $\frac{dT}{dt} = -k(T - T_e)$

**c.** $\frac{dm}{dt} = -km$

**d.** $\frac{dN}{dt} = kN(S - N)$

**30** [Intermediate] • Target: Solving by separation of variables

Find the general solution of $\frac{dy}{dx} = 2x^2$.

**a.** $y = \frac{2x^3}{3} + C$

**b.** $y = \frac{2x^3}{3}$

**c.** $y = 6x^3 + C$

**d.** $y = 2x^2 + C$

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**31** [Intermediate] • Target: Forming differential equations from a description

A quantity satisfies $\frac{dy}{dt} = -y$ with $y(0) = 9$. Find $y$ when $t = \ln(3)/1$.

**a.** 4

**b.** 27

**c.** 3

**d.** 9

**32** [Intermediate] • Target: Solving modeling equations with initial conditions

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

**a.** -1

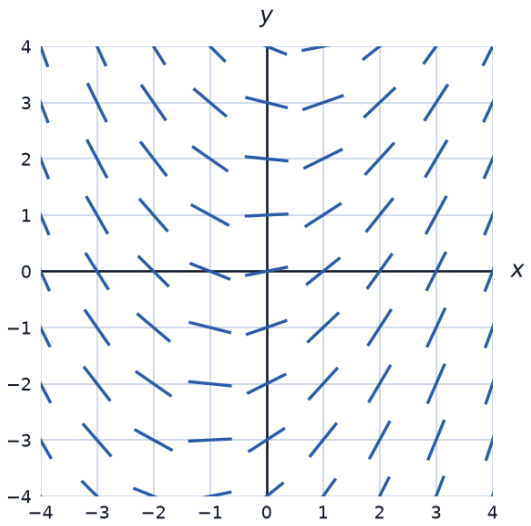
**b.** 1

**c.** $1 + e^2$

**d.** 2

**33** [Intermediate] • Target: First-order linear differential equations using an integrating factor

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.




**a.** $e^{-1} + 1$

**b.** $2e^{-1} + 1$

**c.** $e^{-1} - 1$

**d.** $2 - e^{-1}$

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**34** [Intermediate] • Target: Further integrating factor examples

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

**a.** $20 + 60$

**b.** $60e^{-1}$

**c.** $80e^{-1}$

**d.** $20 + 60e^{-1} \approx 42.1$

**35** [Intermediate] • Target: Verifying solutions to differential equations

A second-order ODE with repeated root $r = r_1$ has general solution:

**a.** $y = C_1e^{r_1x} + C_2e^{-r_1x}$

**b.** $y = C_1e^{r_1x}$

**c.** $y = (C_1 + C_2)e^{2r_1x}$

**d.** $y = (C_1 + C_2x)e^{r_1x}$

**36** [Intermediate] • Target: General and particular solutions

A 200-L tank holds brine with 20 kg of salt. Brine with $\frac{1}{2}$ kg/L enters at 10 L/min and the well-mixed solution leaves at the same rate. How much salt (kg) is in the tank after 20 min?

**a.** $100 - \frac{80}{e}$

**b.** $100 - \frac{100}{e}$

**c.** $120$

**d.** $\frac{20}{e} + 100$

**37** [Intermediate] • Target: Applications: Newton's law of cooling

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

**a.** $\tan x$

**b.** $\ln|\cos x|$

**c.** $-\ln|\cos x|$

**d.** $-\cos x$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: A well-designed word problem: a mixing tank • Cross-Lesson: 15.01 Hypothesis Testing

Integrating factor for $y' + P(x)y = Q(x)$:

**a.** $e^{P(x)}$

**b.** $e^{\int P\,dx}$

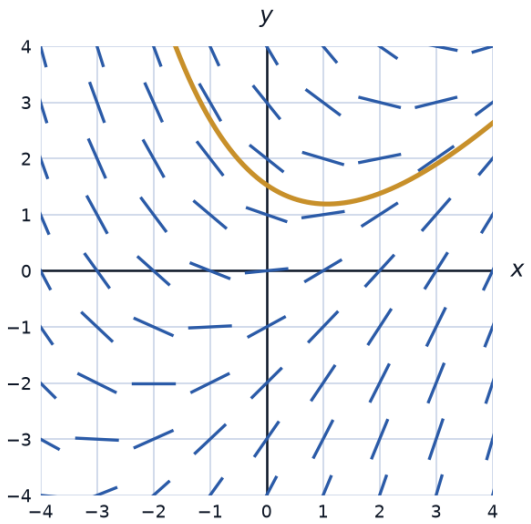
**c.** $P(x)$

**d.** $1/P(x)$

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**39** [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 2.01 Exponentials & Logarithms

Initial value problem: $y' = y$, $y(0) = 2$. Solution:




**a.** $y = e^{2x}$

**b.** $y = 2e^{x}$

**c.** $y = 2x + 2$

**d.** $y = xe^{x}$

**40** [Challenging] • Target: Forming differential equations from a description • Cross-Lesson: 3.01 Advanced Trigonometry

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

**a.** $e^{i\beta x}$

**b.** $C_1 e^{\alpha x} + C_2 e^{\beta x}$

**c.** $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

**d.** $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

**41** [Challenging] • Target: Solving modeling equations with initial conditions • Cross-Lesson: 4.01 Differentiation

Fundamental theorem for linear ODEs: the general solution of $Ly = g$ is:

**a.** $y_p$ only

**b.** $y_h + y_p$

**c.** $y_h$ only

**d.** any solution of $Ly = 0$

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**42** [Challenging] • Target: First-order linear differential equations using an integrating factor • Cross-Lesson: 5.01 Integration

The first, fourth, and sixth terms of an arithmetic progression $(u_n)$ are equal to the first, second, and third terms, respectively, of an infinite geometric progression $(v_n)$ with common ratio $re1$. Given that the sum to infinity of the geometric progression is $27$, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first $n$ terms of the arithmetic progression.

**a.** $44$

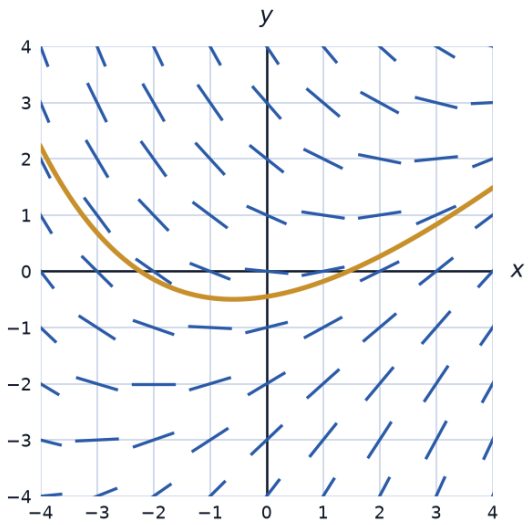
**b.** $45$

**c.** $50$

**d.** $54$

**43** [Challenging] • Target: Further integrating factor examples • Cross-Lesson: 6.01 Sequences & Series

The first three terms of a geometric progression are given by $u_1 = 3k - 2, \quad u_2 = k + 2, \quad u_3 = k - 2,$ where $k$ is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, $S_\infty$.




**a.** $32$

**b.** $-1$

**c.** $\frac{32}{3}$

**d.** $16$

**44** [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 7.01 Vectors

The continuous random variable $X$ has probability density function given by

$f(x) = \{k(4 - x^2), 0 \le x \le 2, ; 0, \text{otherwise},$ where $k$ is a constant. Find the value of $k$.

**a.** $\frac{3}{16}$

**b.** $\frac{1}{4}$

**c.** $\frac{3}{8}$

**d.** $\frac{16}{3}$

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**45** [Challenging] • Target: General and particular solutions • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The first three terms of a geometric progression are $32$, $24$, and $18$. Find the sum to infinity, $S_\infty$, of the progression.

**a.** 128

**b.** 96

**c.** 74

**d.** 64

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